Appendix for

Ambient Air Pollution and the Transmission of Infectious Diseases

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Appendix A: Data Appendix for COVID-19 Analysis

1. COVID-19 data

The COVID-19 data are retrieved from the China National Health Commission (CNHC).* The data comprise newly infected, recovered, and death cases from 20 January to 1 April 2020 in 330 prefectural cities in China (See Figure 1 and Appendix Table 2 for summary statistics). During this period, there were 49,982 cases and 2,553 deaths in Wuhan and 30,441 cases and 727 deaths in other cities. By 1 April 2020, COVID-19 was largely controlled in the country and more than 95% of confirmed cases recovered (tested as negative), so we observed the entire cycle of the virus transmission (See Figure 2). In our baseline analyses, we exclude Wuhan city due to concerns about the city's data quality (Q. Li et al., 2020).†

2. Air quality data

The air quality data are obtained from 1,605 air quality monitoring stations covering all of the prefectural cities in China. These data are collected from China's National Urban Air Quality Real-Time Publishing Platform.‡ We use the Air Quality Index, which is a comprehensive measure of air pollution: the index is constructed using $PM_{2.5}$, PM_{10}, SO_2, CO, O_3, and NO_2 concentrations, with a lower AQI meaning better air quality. In China, the AQI is determined by the maximum concentration of different air pollutants. To create the city-level air quality data, we first calculated the distance from a city's population center to all monitoring stations within the corresponding city. We then aggregated station-level air pollution data to city-level data using inverse distance weights.

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* China National Health Commission (accessed 1 July 2020); http://en.nhc.gov.cn/index.html

† Wuhan was the epicenter of COVID-19 in China. During the first few weeks after the COVID-19 outbreak, the city faced severe medical resource shortages, and many patients could not get immediate diagnosis and treatment. On 17 April, the Chinese government added another 1,290 COVID-19 deaths in Wuhan (without reporting the timing of the fatalities). COVID-19 data outside Wuhan do not suffer from similar problems because there were far fewer COVID-19 cases in these cities, and COVID-19 tests became widely available soon after scientists learned about the situation in Wuhan. However, our results remain robust when we include data from Wuhan.

‡ Data can be assessed through the following website: http://106.37.208.233:20035. A recent study suggests that air quality data's accuracy improved substantially when automatic air pollution monitoring was introduced from 2013 to 2015 (Greenstone et al., 2020)

For this process, stations closer to the population center are given higher weights so that city-level air pollution data can better represent the population of each city. The weights are inversely proportional to the square distance.

2. Thermal inversion data

The thermal inversion data are obtained from the MERRA-2.[§] The data include the temperature in 42 atmospheric layers (110m to 36,000m). Following (Arceo et al., 2016b; He et al., 2019), we use the difference in temperature between the first layer (110m) and the second layer (320m) because this is expected to be associated with air pollution at the ground level. The raw data includes the information for each 50*60 km grid. We aggregate the grid-level data to the city level using the same methodology as the air pollution data.

3. Weather data

Weather data include temperature, precipitation, and snow depth. These data were obtained from the Global Historical Climatology Network (GHCN) from the U.S. National Oceanic and Atmospheric Administration (NOAA).[**] We collapse these data to the city-by-day level using the same method as the air quality data.

Appendix B: The Effect of Air Pollution on the COVID-19 Death Rate

Existing studies suggest that air pollution increases deaths from COVID-19, likely because air pollution could impair the physical capability to recover from the infection. This is policy-relevant, but our dataset does not have sufficient statistical power to investigate the relationship. Outside Wuhan, there were only around 700 deaths from the pandemic in China, with about 90% of cities having no death or just one death. While more than 3,000 deaths are recorded in Wuhan, the data

[§]MERRA-2 (National Aeronautics and Space Administration, Goddard Space Flight Center, accessed 1 July 2020); https://gmao.gsfc.nasa.gov/reanalysis/MERRA-2/

[**] Global Historical Climatology Network (National Oceanic and Atmospheric Administration, accessed 1 July 2020); https://go.nature.com/2Vy9fez

may not be accurate. For example, on 17 April 2020, the official COVID-19 death record in Wuhan was revised, with 1,290 deaths being added but without telling us the timing of the deaths.

Nevertheless, here we examine the relationship between air pollution and death rate by aggregating the daily level data to the weekly level so that we can have a wider variation in the outcome. Our outcome variable is not the growth rate of deaths because it should be proportional to the growth rate of the infections. Instead, we use the death rate, which is defined as the probability of death given infection. Using this outcome variable, we fit the following model, analogous to equation (7),

$$\rho_t = \frac{D_t - D_{t-1}}{I_{t-1}} = \frac{new\ deaths_t}{active\ infection_{t-1}} = \rho_{t0} + f(AP_{it}) + \varepsilon_{it} \qquad (A1)$$

where the death rate is a function of baseline death rate, air pollution, and mean zero error term. We use the same control variables and a set of fixed effects as in equations (13) – (15).

We do not find any suggestive evidence that air pollution increases the death rate of the patients (Appendix Figure 6). All coefficients seem to be very small, and the sign is not consistent. As expected, our data may not have sufficient statistical power to investigate the relationship accurately.

Appendix C: Back-of-the-envelope calculation

We estimate the excess COVID-19 cases attributable to poor air quality. Developing countries, including China, have suffered greatly from poor environmental quality at the cost of rapid economic growth (Greenstone et al., 2020). In China, AQI lower than 100 is regarded as "moderate" or "good" air quality and is recognized as the "blue sky." However, during our study period, 18.6% of the samples (city-by-day) did not meet this standard, and 30% of the cities explain 72.9% of the share with poor air quality. To estimate the excess COVID-19 cases, we first estimate their daily growth rate in two scenarios: observed air quality and the blue sky scenario (AQI is always lower

than 100) by calculating the following equations.

$$\hat{g}_{it}^{b} = g_{0it} - \sum_{k} (AQI_{i,t-k} - AQI_{i,t-k}^{b}) * f(\hat{\alpha}_{t-k}^{z}) \quad (A2)$$

where g_{0it} denotes the observed daily growth rate in city *i* at time *t*, and g_{it}^{b} denotes the growth rate if the AQI were always lower than 100. $AQI_{i,t-k}$ is observed AQI and $AQI_{i,t-k}^{b}$ is the hypothetical AQI, in that AQI higher than 100 is replaced with 100. $f(\hat{\alpha}_{t-k}^{z})$ represents the estimated coefficients from our main estimates (equation (15)). As discussed in the main analyses, we only consider impacts between k = 2 and k = 13.

Using the predicted growth rate, the active infections in the blue sky scenario would have been:

$$\widehat{active}_{it}^{b} = \widehat{active}_{i,t-1}^{b} * \exp(\hat{g}_{it}^{b}) \quad (A3)$$

The predicted number of active infections in city *i* at time *t* builds on that number in the previous period. To predict these numbers, we use the initial active cases in each city when it exceeds 10. We then use the observed removal rate in equations (3) and (4) to estimate the cumulative confirmed cases.

For the removal rate, we use the following equation.

$$removal\ rate_{t} = \gamma_{t} + \rho_{t} = \frac{recovered\ individual + dead\ individual\ at\ time\ t}{infections\ at\ t-1} \quad (A4)$$

For this projection, we allow the removal rate to vary over time because it should vary depending on the stage of the outbreak. Note that the individuals are classified as recovered when they are tested as negative in our dataset. Therefore, its definition is slightly different from that in the classical SIRD model.

Appendix D: Data for Influenza Analysis

1. Influenza data

The influenza data were retrieved from the Shimane Prefecture Infectious Disease Surveillance

Center,†† which summarizes the data published by the Ministry of Health, Labour and Welfare.‡‡ The data contain the weekly number of influenza detection reports from approximately 5,500 representative hospitals across 47 prefectures. The data refer to the weekly average number of influenza hospital visits per hospital from the influenza sentinel medical institutions (approximately 5,000 internal medicine and pediatric medical institutions) and the core sentinel medical institutions (approximately 500 medical institutions with over 300 beds as well as internal medicine and surgery departments), covering 47 prefectures in Japan for the period 2010 to 2018. It should be noted that our dataset does not include recovered or deceased cases; therefore, we use confirmed cases for analysis. We use the data from the last week of 2015 to obtain the number of hospitals in each prefecture,§§ and then back-calculate the estimated total number of reports. During the sample period, there were, on average, 1.4 million detection reports annually.

2. Air quality data

The air quality data are obtained from nearly 1,500 air quality monitoring stations covering all the prefectures in Japan. The data are collected from the Japan National Institute for Environmental Studies (JNIES).*** We use the concentration of pollutants to measure air pollution, with a lower value indicating better air quality. In Japan, the Suspended Particulate Matter (SPM), containing microscopic solids or liquid droplets that are so small (less than 10 $\mu g/m^3$) that they can be inhaled, is mainly used to measure prefecture's air pollution because fewer monitoring stations report PM2.5. To create the prefecture-level air quality data, we first calculated the distance from a prefecture's population center to all monitoring stations within the corresponding prefecture. We then aggregated station-by-day level air pollution data to prefecture-by-day level data using inverse square

†† https://www1.pref.shimane.lg.jp/contents/kansen/inf/23_24/inf0004.htm
‡‡ https://www.mhlw.go.jp/stf/seisakunitsuite/bunya/kenkou_iryou/kenkou/kekkaku-kansenshou/infulenza/index.html
§§ https://www.mhlw.go.jp/file/06-Seisakujouhou-10900000-Kenkoukyoku/0000200664.pdf
*** Japan National Institute for Environmental Studies; https://www.nies.go.jp/igreen/tm_down.html

distance weights. Then, we aggregate the daily pollution data to the weekly level by averaging the daily pollution data within one week.

3. Thermal inversion data

We use the same data as the COVID-19 analysis.

4. Weather data

Weather data include temperature, precipitation, and wind speed. These data were obtained from the Meteorological Agency of Japan.[11] The micro weather information is collected by the Automated Meteorological Data Acquisition System, which consists of 1,300 real-time weather stations covering all of the prefectures in Japan. We collapse these data to the prefecture-by-day level using the same method as the air quality data. Then, we aggregate the daily weather data to the weekly level by averaging the daily weather data within one week.

Appendix E: Methods for Influenza Analysis

Based on our empirical framework, we use a similar 2SLS procedure to estimate the IV model:

$$\widehat{SPM}_{i,t-k} = \sum_j Days\ Inversion_{i,t-j} * \delta_{t-j} + \sum_j X_{i,t-j} * \chi_j + \tau_i + \varphi_t + e_{i,t} \quad (18)$$

$$g_{it} = \log(C_{it}) - \log(C_{i,t-1}) = \sum_k \widehat{SPM}_{i,t-k} * \alpha_{t-k} + \sum_k X_{i,t-k} * \sigma_k + \theta_i + \pi_t + \varepsilon_{i,t} \quad (19)$$

where *it* represents prefecture *i* at time (year-by-week) *t*. $Days\ Inversion_{i,t-j}$ denotes the weekly number of inversion days in prefecture *i* in year-by-week *t-j*, defined as days with a positive temperature difference between the first and second layers. We then regress the weekly growth rate of confirmed influenza cases on the predicted pollution concentration ($\widehat{SPM}_{i,t-k}$) from inversion days. We add temperature, precipitation, and wind speed as controls, as well as prefecture fixed effects (τ_i and θ_i) and year-by-week fixed effects (φ_t and π_t). We cluster the standard errors at the

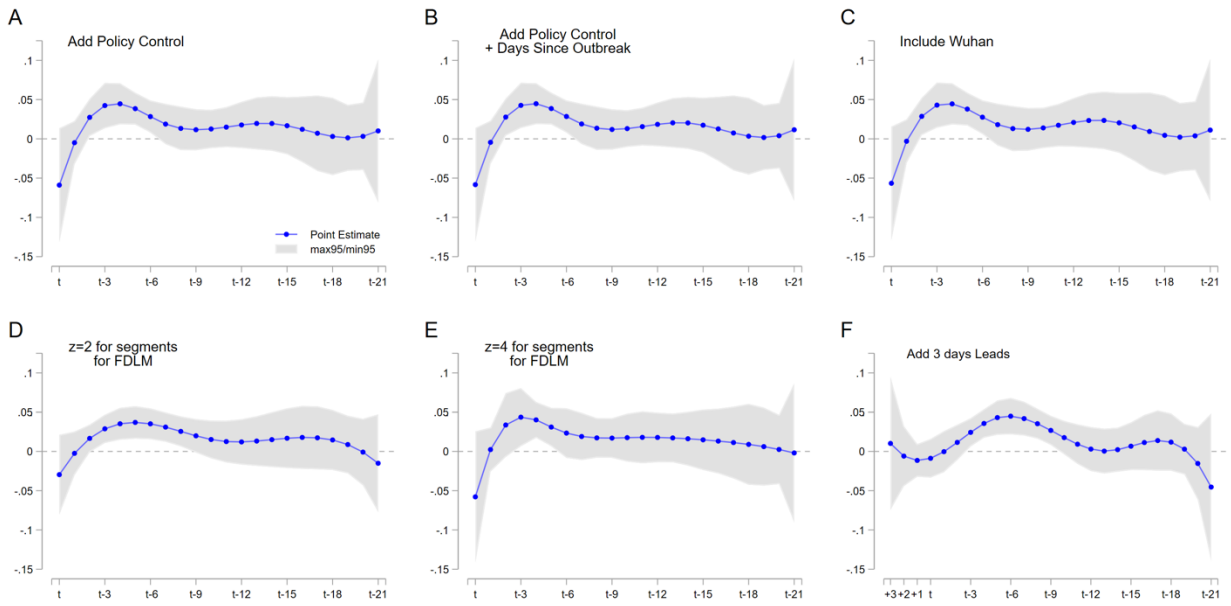
[11] Meteorological Agency of Japan; https://www.jma.go.jp/jma/en/Activities/amedas/amedas.html

prefecture year-by-week level.

We have to modify our base framework in two ways to estimate the pollution-influenza relationship in Japan. First, we use confirmed cases to construct our growth rate. Second, we estimate a distributed lag model, which includes lagged effects for up to two weeks, rather than using the "Flexible" model. This adjustment is necessary because influenza data in Japan are only available at the weekly level, making it challenging to estimate flexible lagged coefficients using polynomial approximation. We focus on the contemporaneous impact of air pollution because the flu incubation period is typically shorter than that of COVID-19, ranging from 1 to 4 days, with symptoms developing 2 to 4 days after infection. Nevertheless, we also estimate the cumulative impacts up to two weeks prior.

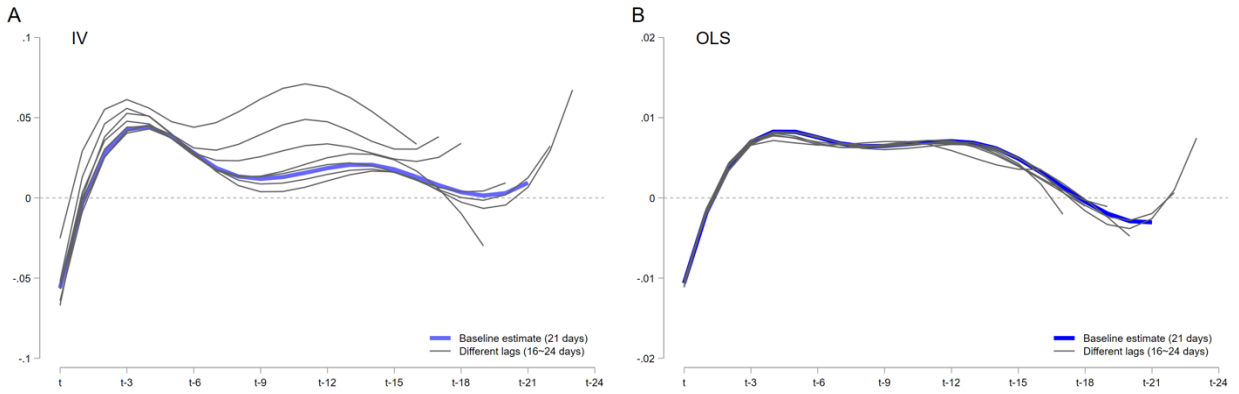
Appendix Figures

Appendix Figure 1. Alternative specifications: air pollution and COVID-19 growth rate



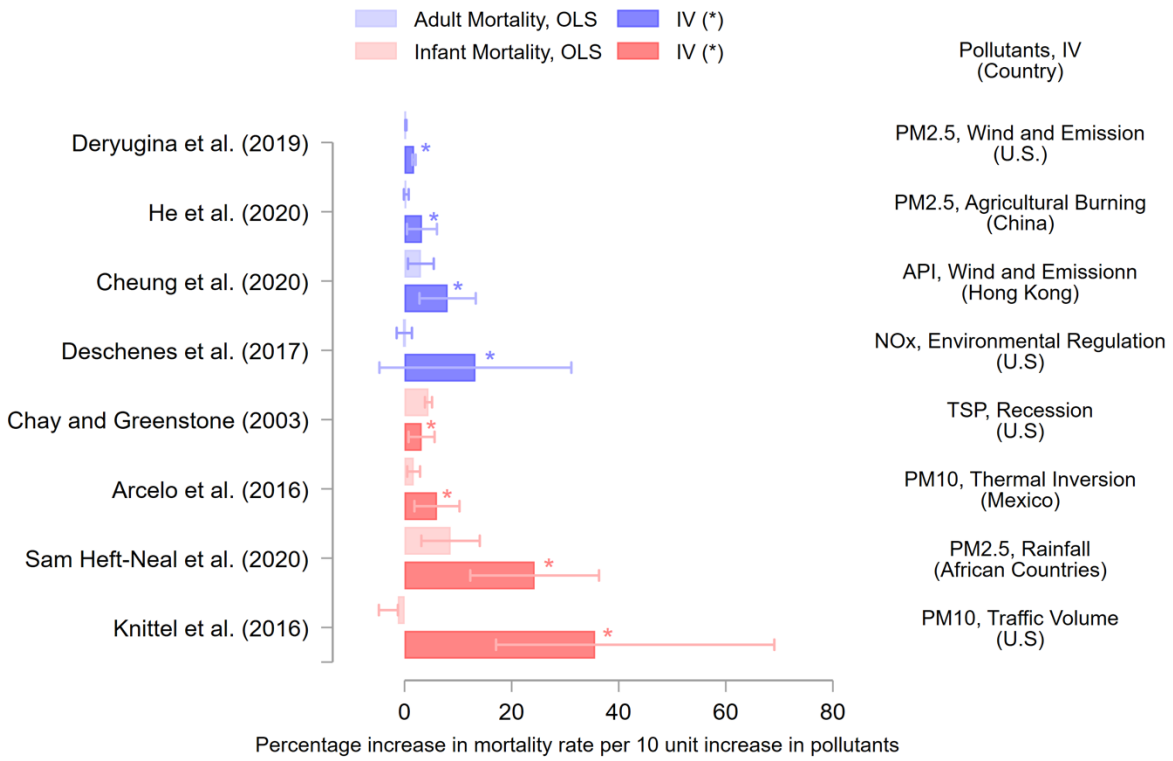
Notes: In Panel A, lockdown status is added as a control variable. In Panel B, the lockdown status and days since the outbreak (the first case confirmed) are added to the regression. In Panel C, we include Wuhan. In Panels D and E, we adopt different segments for the Flexible Distributed Lag Model. In Panel F, we add up to three days of leads of daily air pollution levels in the regression. The joint coefficient for the three leads is -0.002, with the standard error at 0.024, suggesting that future air pollution does not affect the disease growth rate. Thermal inversions and the same controls are used for all first-stage regressions. All regressions include weather controls (temperature, precipitation, wind speed, and snow depth), date fixed effects, and city fixed effects. Standard errors are clustered at the city level.

Appendix Figure 2. Different lags: air pollution and COVID-19 growth rate



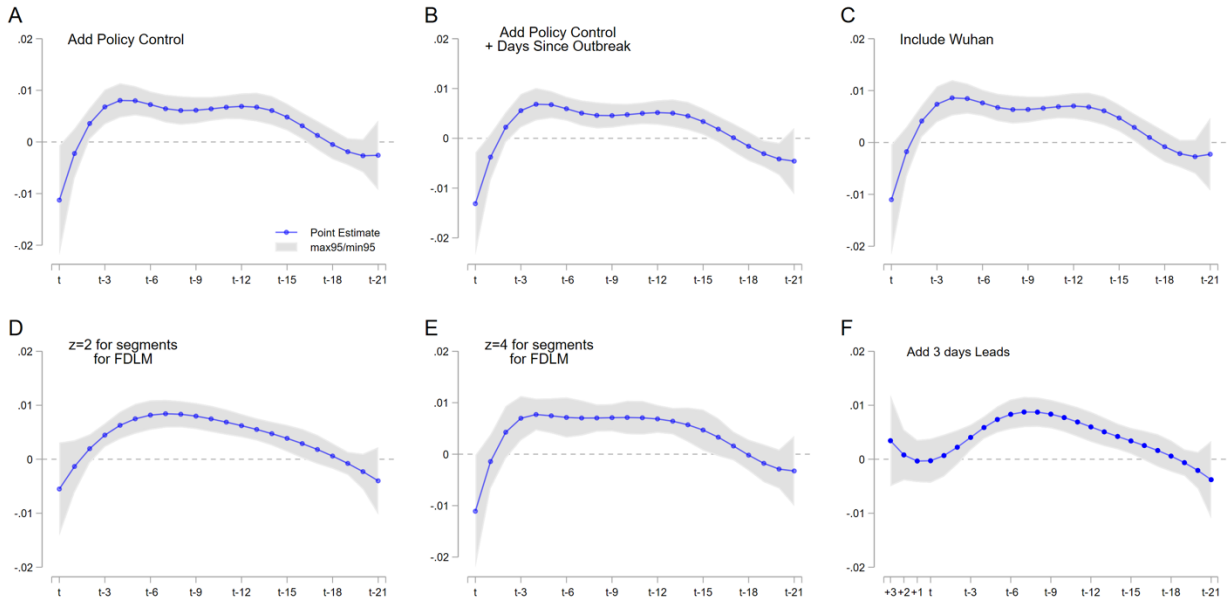
Notes: Panel A represents the IV estimates. We use different lengths of lags from 16 days to 24 days. The blue line plots the baseline estimates (21 days), and the gray line shows the results with different lags. Panel B shows the results using OLS estimates.

Appendix Figure 3. Comparison between IV estimates and OLS estimates



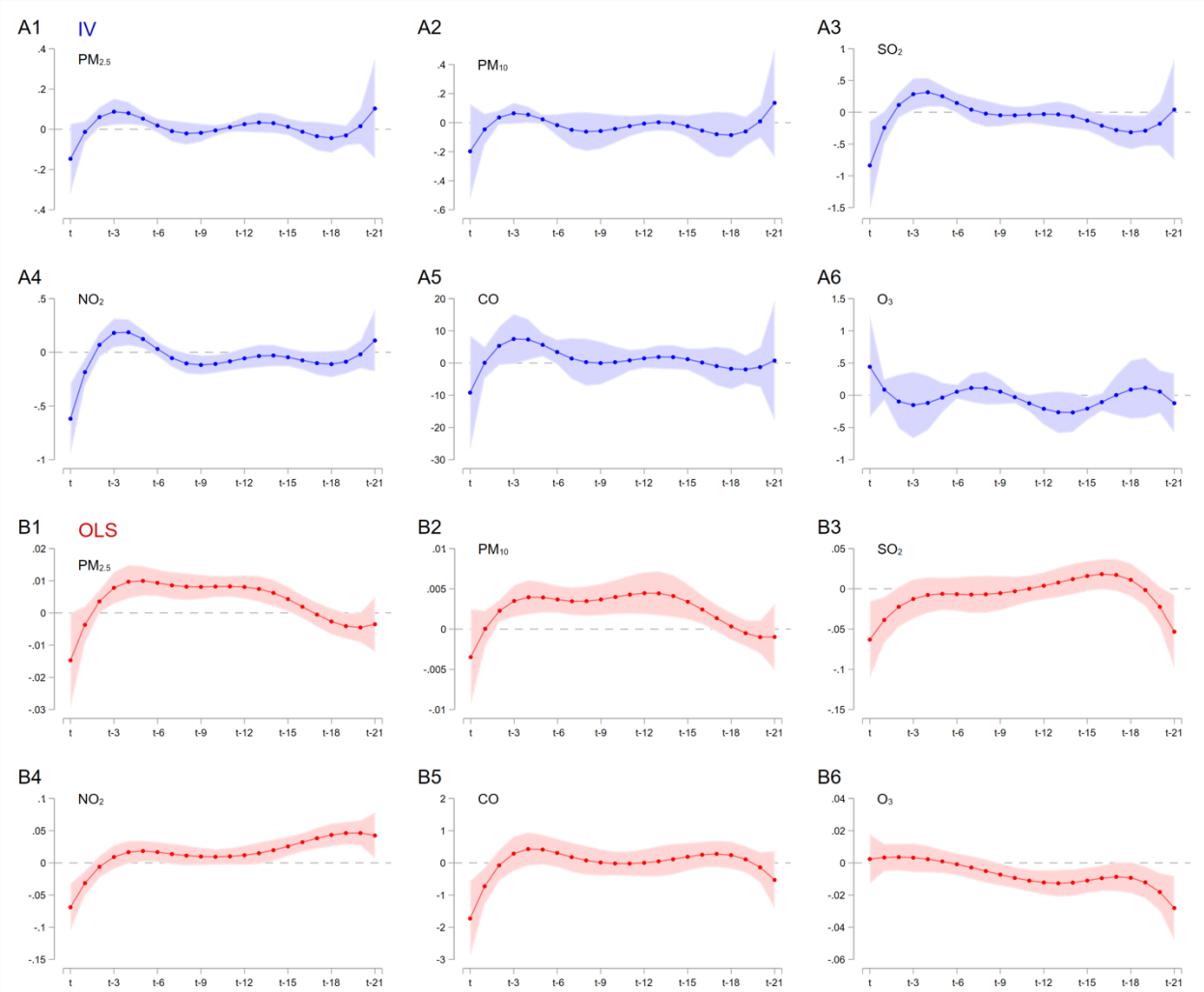
Notes: This figure plots the estimates of the effect per 10 unit increase in air pollution on mortality rate (%). Deryugina et al. (2019) use mortality among those aged above 65. Except for Chay and Greenstone (2003), existing studies report the OLS estimates are smaller than the IV estimates. Note that different studies focus on different pollutants and use different instrumental variables. Therefore, we do not compare estimates across different studies but across different methods within each study.

Appendix Figure 4. OLS estimates: air pollution and COVID-19 growth rate



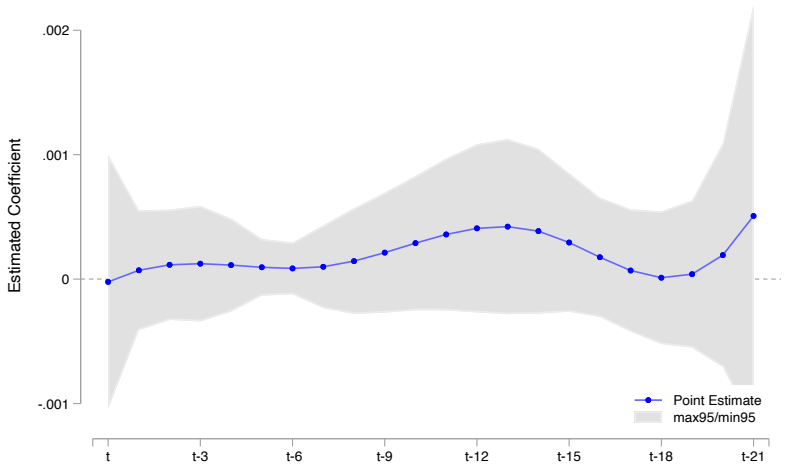
Notes: In Panel A, we include the lockdown status as a control variable. In Panel B, we include the lockdown status and days since the outbreak (the first case confirmed) as control variables. In Panel C, we include data from Wuhan in the regression. In Panels D and E, we use different segments for the Flexible Distributed Lag Model. In Panel F, we add three days of leads daily air quality levels in the regression. The joint coefficient for the three leads is 0.010, with the standard error at 0.010, suggesting that leads of air pollution levels do not affect the disease growth rate. All regressions include weather controls (temperature, precipitation, wind speed, and snow depth), date fixed effects, and city fixed effects. Standard errors are clustered at the city level.

Appendix Figure 5. Air pollution and COVID-19 growth rate: different air pollutants



Notes: Panels A1 to A6 plot the IV estimates. Each figure shows the coefficient of a 1 point increase in each pollutant in each day. Weather controls (temperature, precipitation, and snow depth), date fixed effects, and city fixed effects are included in both the first and second stage regression. The blue dots represent the point estimates, while the blue areas denote the 95% confidence intervals. Panel Bs plot the OLS estimates with the same set of controls. The red dots represent the point estimates, while the red areas denote the 95% confidence intervals. In all regressions, standard errors are clustered at the city level.

Appendix Figure 6. Impacts of air pollution on the COVID-19 death rate



Notes: This figure shows estimated coefficients using IV estimator. We do not find a statistically significant relationship between AQI and the death rate.

Appendix Tables

Appendix Table 1. Summary Statistics for COVID-19 analysis

Variable	Obs	Mean	Std. Dev.	Min	Max
A. COVID-19					
Confirmed Cases (C_t)	30,268	55.6	238	0	3518
Recovered Cases (R_t)	30,268	36.5	179	0	3389
Deceased Cases (D_t)	30,268	1.09	7.86	0	128
Active Cases ($I_t = C_t - R_t - D_t$)	30,268	18	109	-1	2760
Growth Rate (%) ($g_t = \ln(I_t + 1) - \ln(I_{t-1} + 1)$)	29,928	0.19	20.8	-289	391
B. Air Quality					
Air Quality Index	29,701	69.7	49.7	9.08	500
$PM_{2.5}$ (μg/m3)	29,705	46.3	39.7	1.83	1324
PM_{10} (μg/m3)	29,705	70.4	69.3	3.33	2342
SO_2 (μg/m3)	29,705	11.6	9.3	1	129
NO_2 (μg/m3)	29,705	24.7	15	1.67	115
CO (mg/m3)	29,705	0.869	0.449	0.1	6.26
O_3 (μg/m3)	29,701	53.6	20.9	1.55	200
C. Weather					
Thermal Inversion (℃)	29,029	0.864	1.23	0	10.2
Temperature (℃)	30,268	5.95	8.52	-30.7	28.9
Precipitation (mm)	30,268	23.3	61.4	0	1498
Wind Speed (m/s)	30,268	2.43	1.00	0.551	10.378
Snow (depth, mm)	30,268	54.9	31.9	10	412

Appendix Table 2. Summary Statistics for Influenza Analysis in Japan

Variable	Obs	Mean	Std. dev.	Min	Max
A. Influenza					
Influenza	21,996	573	1,513	0	22,225
Growth Rate (%) ($g_t = \ln(I_t + 1) - \ln(I_{t-1} + 1)$)	21,949	-0.026	69	-637	502
B. Air Quality					
SPM (μg/m^3)	22,043	18	7.6	2.3	77
$PM_{2.5}$ (μg/m^3)	19,898	13	5.5	-0.2	55
SO_2 (ppb)	22,043	1.9	1.4	0	12
NO_2 (ppb)	22,043	8.9	4.6	0.77	43
CO (0.1ppm)	11,484	2.9	1.1	0.006	12
O_3 (ppb)	22,043	30	9.2	5.6	65
C. Weather					
Inversion days	22,043	0.16	0.56	0	7
Temperature (℃)	22,043	15	8.7	-11	32
Precipitation (mm)	22,043	34	44	0	583
Wind Speed (m/s)	22,043	17	5.7	4.7	79
Snow (mm)	22,043	0.52	3.0	0	67

Appendix Table 3. Robustness Checks for Influenza Analysis in Japan

Dependent Variable:	OLS			IV		
Weekly disease growth rate (%)	(1)	(2)	(3)	(4)	(5)	(6)
SPM ($\mu g/m^3$)	-0.083	0.017	0.13	3.48	1.56	3.52
	(0.10)	(0.10)	(0.089)	(3.48)	(2.18)	(2.69)
Observations	21,949	21,902	10,951	21,949	21,902	10,951
Prefecture FE + Year-week FE		Yes	Yes		Yes	Yes
Flexible FEs	Yes			Yes		
Add Lead Pollution		Yes			Yes	
Restrict flu season			Yes			Yes
First-stage F-statistics				36.3	59.8	38.8

Notes: This table reports the OLS and IV estimates of the pollution-transmission relationship for influenza. The dependent variable is the weekly growth rate of influenza, measured as the difference in number of detection cases (in logarithm) between the current and last weeks. SPM denotes the concentration of suspended particulate matter. Significance levels are indicated by *** $p<0.01$, ** $p<0.05$, and * $p<0.01$.